SYNOPSIS OF THE PROJECT

Title of the Project :

Furnishers - The Furniture Shop

Introduction:

Furniture stores are an integral part of the retail market, catering to customers seeking stylish and functional home or office furnishings. However, managing the operations of a furniture shop often involves multiple challenges, including tracking inventory, handling customer orders, managing supplier relationships, and processing sales efficiently. Manual management of these tasks can be time-consuming, error-prone, and inefficient, particularly as the business scales.

The project **"Furnishers - The Furniture Shop"** is a desktop-based application designed to address these challenges by automating and digitizing the core functions of a furniture store. Developed using **VB.NET**, this application provides a comprehensive solution for managing various aspects of shop operations, from inventory tracking to sales processing. It replaces traditional manual methods with an intuitive and user-friendly system that minimizes errors, saves time, and enhances productivity.

In addition to its operational benefits, the system is designed to improve the customer experience by enabling faster order processing and personalized service. By maintaining detailed records of customer purchases and preferences, the shop can offer tailored recommendations and promotions, fostering customer loyalty and satisfaction. The software also includes powerful reporting and analytics tools, empowering shop owners to make data-driven decisions to optimize their business operations and boost profitability.

With features like user role management and a secure login system, the application ensures data security and role-based access, safeguarding sensitive business information. The system is adaptable to the specific needs of furniture shops, making it a valuable asset for businesses of all sizes, whether a small boutique store or a larger retail outlet.

In summary, **"Furnishers - The Furniture Shop"** is a modern solution designed to transform the way furniture shops operate, offering convenience, efficiency, and enhanced customer service. It bridges the gap between traditional shop management and contemporary digital solutions, setting the stage for a more organized and profitable business model.

OBJECTIVES & SCOPE OF THE PROJECT

Objective:

The primary objective of the **"Furnishers - The Furniture Shop"** project is to design and develop a comprehensive software solution that automates and streamlines the operations of a furniture shop. The key objectives include:

Automation of Core Processes: Automate tasks such as inventory management, sales processing, customer records, and supplier interactions to reduce manual effort and errors.

Enhanced Customer Experience: Provide faster and more accurate service, maintain customer purchase history, and facilitate personalized interactions to improve satisfaction and loyalty.

Efficient Inventory Management: Track stock levels in real-time, receive alerts for low-stock items, and ensure optimal inventory levels to avoid overstocking or stockouts.

Simplified Sales Management: Generate and manage invoices seamlessly, apply discounts or promotional offers, and keep accurate sales records for easy tracking.

Comprehensive Reporting: Offer detailed reports on sales trends, inventory performance, and customer behavior to support data-driven business decisions.

Data Security and Access Control: Implement a secure system with role-based access to ensure the protection of sensitive business data.

Scalability and Adaptability: Provide a scalable system that can grow with the business and adapt to the specific needs of furniture shops, regardless of their size.

Specific goals include:

1. Providing secure and reliable user authentication with **Login** and **Sign-Up** functionality.
2. Organizing products into categories such as **Sofa**, **Modern Chair**, **Bed**, **Lamp**, **Minimalistic Decor**, and **Coffee Table**, for easy navigation and management.
3. Enabling efficient **billing** and **sales processing**, with options for cash and UPI payment methods.
4. Implementing a **sales promotion page** to display discounts and offers.
5. Equipping administrators with a **DataGrid** interface to search, update, and delete records seamlessly.
6. Ensuring robust backend operations using **MS Access** with **OLEDB** connectivity for data management.

Scope :

1. User thentication:

- A secure **Login Page** for existing users.
- A **Sign-Up Page** for new users to register with the system.

2. Product Categories:

- A Category Page displaying furniture categories: Sofa, Modern Chair, Bed, Lamp, Minimalistic Decor, and Coffee Table.
- Each category page includes product details like name, description and price.

3. Billing and Payments:

- A Billing Page for processing customer purchases.
- Integration of payment methods: Cash and UPI, with easy selection during checkout.

4. Sales Promotions: A Sales Page to highlight discounts and promotional offers on specific products.
5. Admin Functions: A **DataGrid View** for administrators to:-

- Search for specific records.
- Update or delete product, customer, or order data.
- Manage inventory efficiently.

6. Backend and Database: The backend is powered by **MS Access** using **OLEDB** for secure and fast data transactions. The database stores:

- User credentials (for Login and Sign-Up).
- Product details (category, price).
- Sales and billing records.
- Customer information.

7. Target Audience: Furniture store owners and staff requiring an all-in-one solution for shop management.
8. Limitations:

1. The project focuses on a single-location store and does not include multi-branch support.
2. It does not currently integrate with online platforms for e-commerce functionality.

9. Future Scope:

1. Addition of online shopping features, such as a web or mobile app.
2. Integration with advanced reporting tools for analytics.
3. Multi-store management capabilities.

THEORETICAL BACKGROUND OF PROJECT

The development of the **"Furnishers - The Furniture Shop"** project is based on several theoretical concepts and principles in software engineering, database management, and retail management. This section highlights the theoretical foundation underlying the project.

1. Software Development Principles

The project is built on the principles of structured software development to ensure a robust and scalable system:

Modularity:
The application is designed with modular components, such as login, sign-up, category, billing, and admin functionalities. Each module performs a specific task and interacts with others seamlessly.

Object-Oriented Programming (OOP):
The use of VB.NET emphasizes OOP concepts like encapsulation, inheritance, and polymorphism. These concepts improve code reusability, maintainability, and scalability.

User-Centered Design:
The interface is designed to be user-friendly and intuitive, reducing the learning curve for users like shop owners, sales staff, and administrators.

2. Database Management

The backend of the system is powered by **MS Access**, which is a lightweight and reliable database management system suitable for small to medium-sized applications. The following database principles are applied:

Relational Database Model:
Data is stored in relational tables for efficient storage and retrieval. Tables are designed for entities such as products, customers, sales, and users.

Normalization:
The database is normalized to eliminate redundancy and ensure data integrity. This reduces data anomalies and optimizes storage.

OLEDB Connection:
OLEDB (Object Linking and Embedding Database) is used as the connection interface between the VB.NET application and the MS Access database. OLEDB provides a fast, secure, and efficient way to handle database operations.

3. Retail Management Concepts

The application aligns with standard practices in retail management to meet the operational needs of a furniture store:

Inventory Management:
Tracking product availability is essential to prevent overstocking or stockouts. This project incorporates real-time stock level updates and low-stock alerts.

Sales Processing:
A seamless billing system ensures quick checkout for customers, while discounts and offers promote sales and customer retention.

Customer Relationship Management (CRM):
The system maintains detailed records of customer purchases, enabling personalized recommendations and improved customer service.

4. Secure System Design

Security is a critical aspect of any application that handles sensitive data such as customer details and sales records. The project adheres to the following security practices:

Authentication:
A login system is implemented to ensure that only authorized users can access the application.

Role-Based Access Control (RBAC):
Different user roles (e.g., Admin, Sales Staff) are defined to restrict access to certain features, safeguarding critical business data.

Data Validation:
Input validation mechanisms are included to prevent SQL injection and other malicious activities.

5. Payment System Integration

The project incorporates cash and UPI (Unified Payments Interface) as payment options. The theoretical background for payment processing includes:

Payment Validation:
Ensuring that the payment details provided by the user are accurate and complete before processing.

Digital Payment Systems:
UPI integration simplifies cashless transactions, ensuring secure and instant payment processing.

6. Reporting and Decision Support

The system includes reporting features that assist shop owners in making data-driven decisions:

Sales Analysis:
Generating reports on daily, monthly, and yearly sales trends.

Inventory Insights:
Identifying fast-moving products and those with low demand to optimize inventory management.

Customer Trends:
Analyzing customer purchase patterns to design targeted promotions and loyalty programs.

7. Software Development Methodology

The project follows the **Waterfall Model** of software development, which involves a sequential approach with the following stages:

Requirement Analysis:
Gathering and analyzing the requirements for the furniture shop, such as inventory, billing, and admin functions.

System Design:
Designing the database schema, user interface, and application architecture.

Implementation:
Coding the application using VB.NET and connecting it to the MS Access database using OLEDB.

Testing:
Conducting thorough testing to identify and fix bugs.

Deployment and Maintenance:
Deploying the system at the furniture shop and providing ongoing support for updates and enhancements.

DEFINITION OF PROBLEM

In traditional furniture retail stores, managing inventory, processing orders, and generating invoices manually can be time-consuming and prone to errors. The absence of an automated system results in inefficiencies that impact both business operations and customer satisfaction. The **Furnishers - The Furniture Shop** project aims to address these challenges by developing a structured and digitalized platform for furniture shop management.

Challenges Faced in Traditional Furniture Stores

- **Manual Inventory Management:** Keeping track of product stock manually leads to mismanagement, overstocking, or stock shortages.
- **Time-Consuming Order Processing:**Placing and managing customer orders without an automated system increases the risk of delays and incorrect deliveries.
- **Billing and Invoicing Errors:**Handwritten invoices or manual calculations can result in pricing mistakes, affecting financial accuracy.
- **Limited Customer Interaction:**Without a digital catalog, customers rely on physical store visits, which limits their ability to browse and compare products efficiently.
- **Lack of Sales and Data Analysis:** Without a centralized system, business owners struggle to track sales trends, customer preferences, and inventory turnover.

Need for a Digital Solution

The increasing demand for **automation in retail management** highlights the need for a **centralized and efficient system** that can manage all aspects of a furniture store. A **VB.NET-based desktop application** integrated with **MS Access** can offer:

- **Automated Inventory Management:** Real-time updates on stock availability.
- **Faster Order Processing:** A structured process from product selection to checkout.
- **Accurate Billing System:** Digital invoices eliminate manual errors.
- **User-Friendly Interface:**Customers can browse products conveniently, and administrators can manage store operations efficiently.
- **Data-Driven Business Decisions:** Reports and analytics help store owners optimize inventory and pricing strategies.

The **Furnishers - The Furniture Shop** project aims to overcome these limitations by providing an integrated solution that improves operational efficiency, enhances customer experience, and supports business growth.

SYSTEM ANALYSIS AND DESIGN

System Analysis :

The **Furnishers - The Furniture Shop** application is designed to automate furniture store operations, improving efficiency in inventory management, order processing, and billing. The system is divided into two main user roles: customers and administrators. Customers can browse and purchase furniture products, while administrators can manage inventory, track orders, and generate invoices.

Functional Requirements

- **User Authentication:** Customers and admins must log in to access the system.
- **Product Browsing:** Customers can explore different categories of furniture, including modern sofas, dining tables, and coffee tables.
- **Cart and Checkout:** Customers can add products to their cart and proceed to payment.
- **Billing System:** Automated invoice generation after purchase completion.
- **Admin Dashboard:** Allows administrators to manage product inventory, view orders, and track sales.

Non-Functional Requirements

- **Usability:** The system provides a user-friendly interface for smooth navigation.
- **Performance:** The application ensures fast loading times and efficient data retrieval.
- **Security:** Secure login and data encryption protect user information.
- **Scalability:** The system is designed to accommodate future expansion, such as adding more product categories.
- **Reliability:** Ensures consistent and error-free operations for customers and administrators.

System Design :

System Architecture

The application follows a **three-tier architecture:**

- **Presentation Layer:** The user interface, developed using **VB.NET Windows Forms**, allows customers and administrators to interact with the system.
- **Business Logic Layer:** Handles order processing, inventory updates, and user authentication.
- **Data Layer:** Stores customer details, product inventory, sales data, and invoices in **MS Access** using **OLEDB connectivity**.

Database Design

The **MS Access database** consists of structured tables for managing store operations.

- **Users Table:** Stores customer and admin login details.
- **Products Table:** Contains product details, including name, category, price, and stock quantity.
- **Orders Table:** Records customer orders and payment details.
- **Billing Table:** Stores invoice details for completed purchases.

The **Furnishers - The Furniture Shop** system is designed to ensure **seamless integration between inventory, orders, and billing**, providing an efficient and structured solution for furniture store management..

SYSTEM PLANNING

The **Furnishers - The Furniture Shop** application is designed to streamline furniture store operations by integrating inventory management, order processing, and billing into a single digital platform. The system planning phase involves defining the technology stack, resources, and development methodology required for successful implementation.

Technology Used

- **Frontend:** VB.NET (Windows Forms) for user interface development.
- **Backend:** MS Access as the database for storing inventory, orders, and user information.
- **Database Connectivity:** OLEDB for seamless interaction between the application and the database.
- **Development Environment:** Visual Studio as the primary IDE for coding and debugging.

Material Resources

- **Software Tools:** Visual Studio, MS Access, and OLEDB drivers.
- **Testing Devices:** Computers with Windows OS for running and testing the application.
- **Database Management Tools:** MS Access for handling structured data storage.

Project Timeline

The development process is divided into several phases to ensure systematic progress:

- **Requirement Analysis:** Identifying key features and system functionality.
- **System Design:** Creating database schemas, UI wireframes, and system architecture.
- **Development:** Coding the user interface, business logic, and database integration.
- **Testing:** Conducting unit testing, functional testing, and database validation.
- **Deployment:** Installing the application on client machines and providing user documentation.
- **Maintenance and Updates:** Monitoring performance, fixing bugs, and adding new features based on feedback.

The **Furnishers - The Furniture Shop** project is planned to be a structured and well-organized system, ensuring efficiency in furniture store operations and providing a user-friendly experience for both customers and administrators.

METHODOLOGY ADOPTED, SYSTEM IMPLEMENTATION & DETAILS OF HARDWARE & SOFTWARE USED

METHODOLOGY ADOPTED :

The development of **Furnishers - The Furniture Shop** follows the **Waterfall Model**, ensuring a structured and sequential approach to software development. Each phase is completed before moving to the next, minimizing risks and ensuring proper documentation.

Key phases of the **Waterfall Model** used in this project: -

- **Requirement Analysis:** Identifying business needs, user requirements, and system functionality.
- **System Design:** Creating database structures, UI layouts, and defining business logic.
- **Implementation:** Coding the application using **VB.NET**, integrating **MS Access** as the database, and implementing the **OLEDB connection**.
- **Testing:** Conducting unit testing, functional testing, and validating database integrity.
- **Deployment:** Installing the application and providing user manuals.
- **Maintenance:** Monitoring the system for errors, updates, and feature enhancements.

SYSTEM IMPLEMENTATION :

The implementation process involves a structured development workflow, ensuring the application meets business requirements and user expectations.

- **Phase 1: Requirement Gathering and Planning–** Defining project objectives, identifying key functionalities, and outlining database schema.
- **Phase 2: System Design–** Creating wireframes for UI design, defining database relationships, and establishing system flow.
- **Phase 3: Development –** Writing code for user authentication, inventory management, order processing, and billing.
- **Phase 4: Testing–** Validating system functionality, checking for bugs, and ensuring database security.
- **Phase 5: Deployment–** Installing the software on the client system and providing training for users.
- **Phase 6: Maintenance & Updates–** Performing regular checks, fixing issues, and improving features based on user feedback.

DETAILS OF HARDWARE & SOFTWARE USED :

Hardware Requirements:-

- **Processor:** Intel Core i5 or higher.
- **RAM:** Minimum 4GB (8GB recommended).
- **Storage:** Minimum 250GB HDD or SSD.
- **Display:** Minimum resolution of 1366x768 pixels.
- **Operating System:**Windows 7 or later.

Software Requirements:-

- **Programming Language:** VB.NET for application development.
- **Database:**MS Access for storing inventory, orders, and user data.
- **Database Connectivity:** OLEDB for communication between VB.NET and MS Access.
- **Development Tools:** Visual Studio for writing and debugging code.
- **Testing Tools:** Manual testing and database validation tools.

The **Furnishers - The Furniture Shop** system is developed with an efficient methodology, structured implementation plan, and optimized hardware and software resources, ensuring reliability and ease of use.

SYSTEM MAINTENANCE & EVALUATION

SYSTEM MAINTENANCE

The maintenance phase ensures that **Furnishers - The Furniture Shop** operates efficiently, remains up to date, and continues to meet user requirements. Maintenance includes fixing errors, improving performance, and enhancing features over time.

- **Corrective Maintenance:** Fixing bugs, errors, and any issues reported by users.
- **Adaptive Maintenance:** Updating the system to ensure compatibility with newer versions of Windows and database updates.
- **Perfective Maintenance:** Adding new features such as enhanced reporting, additional payment options, or improved UI.
- **Preventive Maintenance:** Regular database backups, security checks, and system optimizations to prevent future issues.

SYSTEM EVALUATION

Evaluation ensures that the system meets its objectives, operates efficiently, and provides a good user experience.
Key evaluation criteria include:

- **Performance:** Monitoring system speed, data retrieval efficiency, and application responsiveness.
- **User Feedback:** Collecting feedback from store administrators and customers to identify areas for improvement.
- **Feature Usage Analysis:** Tracking how often specific features, such as inventory updates and order processing, are used.
- **Business Impact:** Measuring improvements in sales, order accuracy, and customer satisfaction due to the system's implementation.
- **Security Evaluation:** Ensuring data protection, secure transactions, and restricted access to administrative functions.

Regular **maintenance and evaluation** ensure that the **Furnishers - The Furniture Shop** system remains **efficient, reliable, and scalable**, supporting long-term business growth.

COST AND BENEFITS ANALYSIS

Costs:

The development and implementation of **Furnishers - The Furniture Shop** involve several cost components, including software tools, hardware requirements, and operational expenses.

- **Development Costs:** Expenses related to software development, including coding, testing, and debugging.
- **Hardware Costs:** Computers with necessary specifications for development, testing, and deployment.
- **Software Costs:** Licensing costs for development tools like **Visual Studio** and database software like **MS Access**.
- **Maintenance Costs:** Ongoing updates, security patches, and troubleshooting support.
- **Training Costs:** Educating store employees on how to use the system effectively.

Benefits:

Implementing a **digital furniture shop management system** offers several advantages that outweigh the initial investment.

- **Increased Efficiency:** Automates order processing, inventory management, and billing, reducing manual workload.
- **Error Reduction:** Eliminates calculation errors in invoices and minimizes stock mismanagement.
- **Better Customer Experience:** Enables customers to browse products, place orders, and complete transactions seamlessly.
- **Sales Tracking and Analytics:** Provides business owners with insights into product demand, sales trends, and inventory status.
- **Scalability:** The system can be expanded to support additional stores, product categories, and user roles.
- **Long-term Cost Savings:** Reduces paperwork, optimizes inventory control, and enhances overall business operations.

By analyzing both **costs and benefits,** it is evident that **Furnishers - The Furniture Shop** is a **valuable investment** for furniture retailers, improving efficiency, accuracy, and customer satisfaction while ensuring **sustainable business growth**.

DETAILED LIFE CYCLE OF THE PROJECT

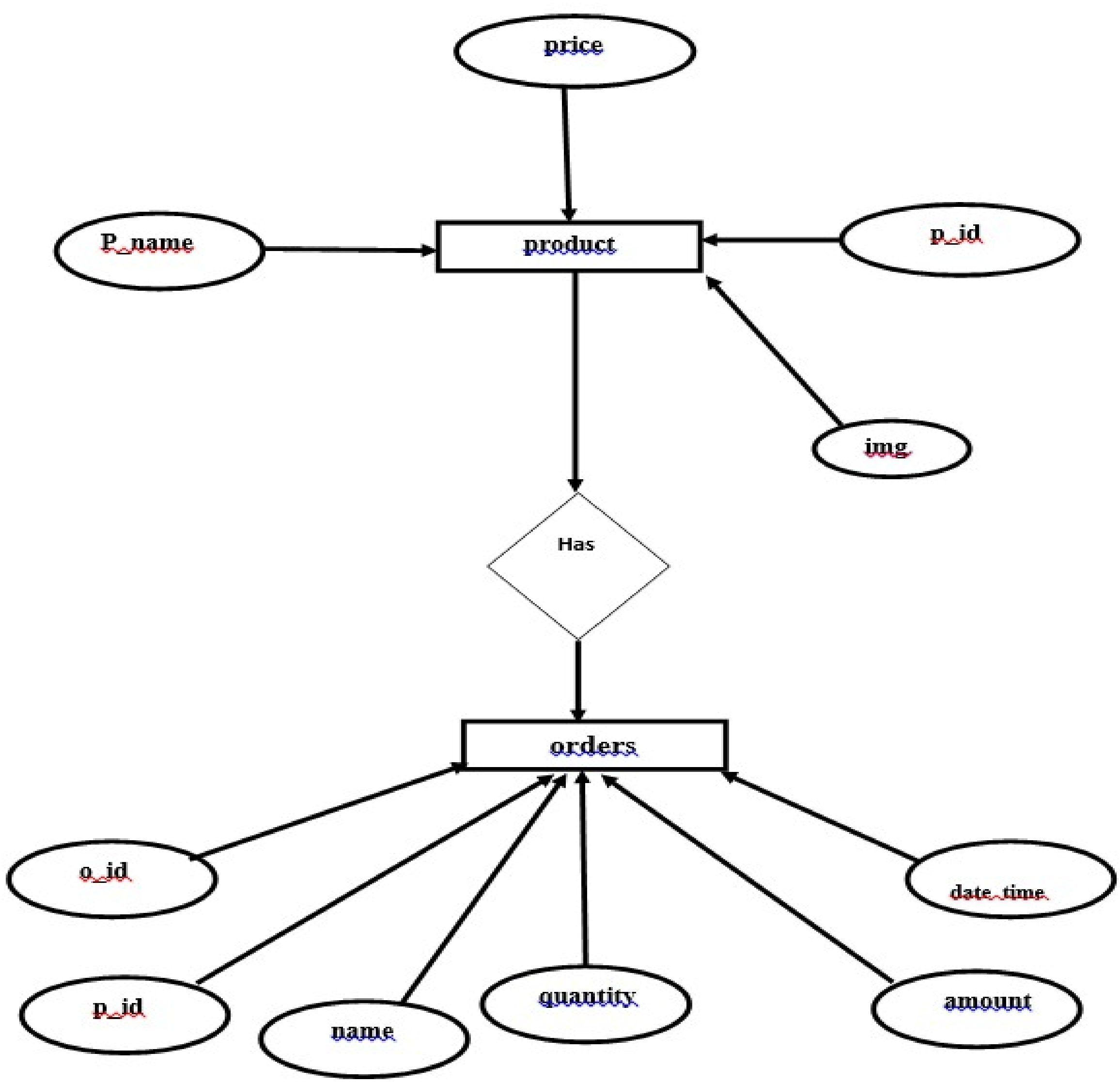
Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) helps represent the database structure and relationships between entities in the system.The **Furnishers - The Furniture Shop** database follows a relational structure, ensuring efficient data management and retrieval.

Signup Table: Stores customer and admin details, including user ID, name, email.

Products Table: Maintains product information such as product ID, category, name, price, stock quantity, and description.

Order Table: Stores invoice details, including bill ID, order ID, total amount, and payment status.

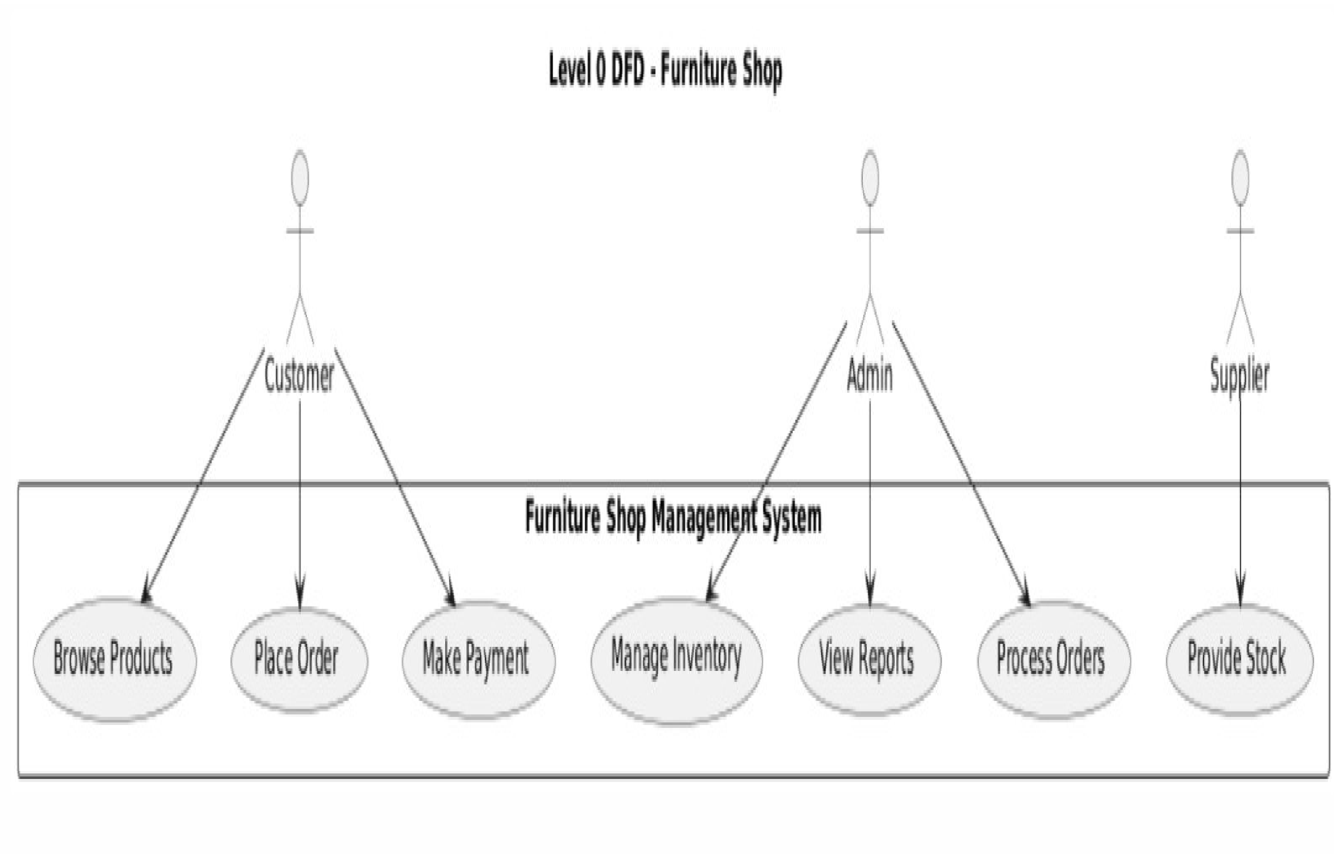


Data Flow Diagram (DFD)

The **Data Flow Diagram (DFD)** illustrates how data moves through the system and the processes involved.

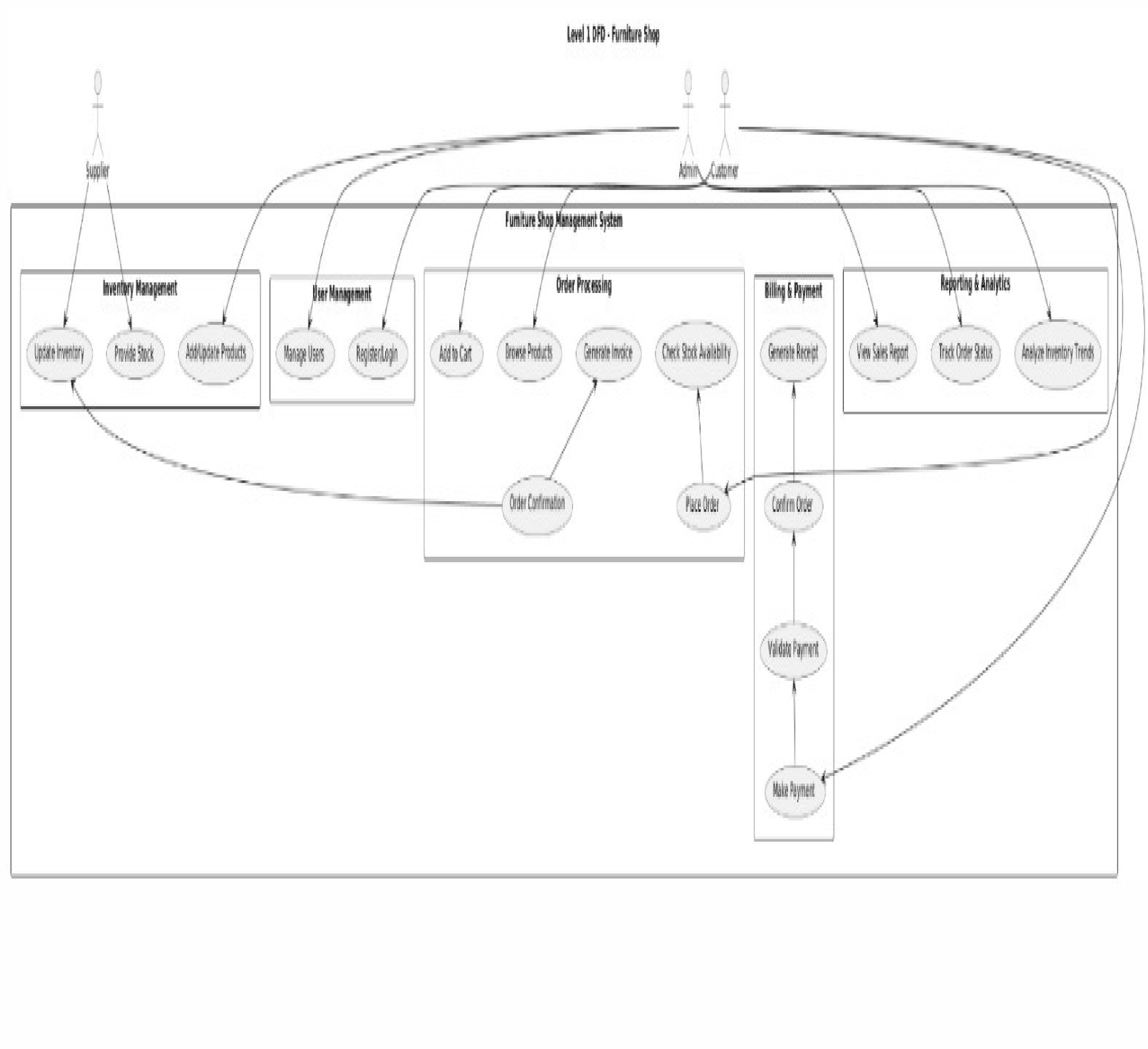
Level 0 (Context Diagram):

- **Customers:** Browse products, place orders, and make payments.
- **Admin:** Manages product inventory, tracks orders, and generates sales reports.
- **System Processes:** Order processing, inventory updates, and billing.
- **Data Stores:** Stores user data, order details, and sales records in MS Access.



Level 1 - Detailed Process Flow:

- Customers log in, view products, add items to the cart, and proceed to checkout.
- Orders are processed, updating stock levels in the database.
- The billing module generates invoices for completed transactions.
- Administrators manage product listings, update inventory, and monitor sales performance.



INPUT AND OUTPUT SCREEN DESIGN

The system features well-structured input and output screens for efficient user interaction.

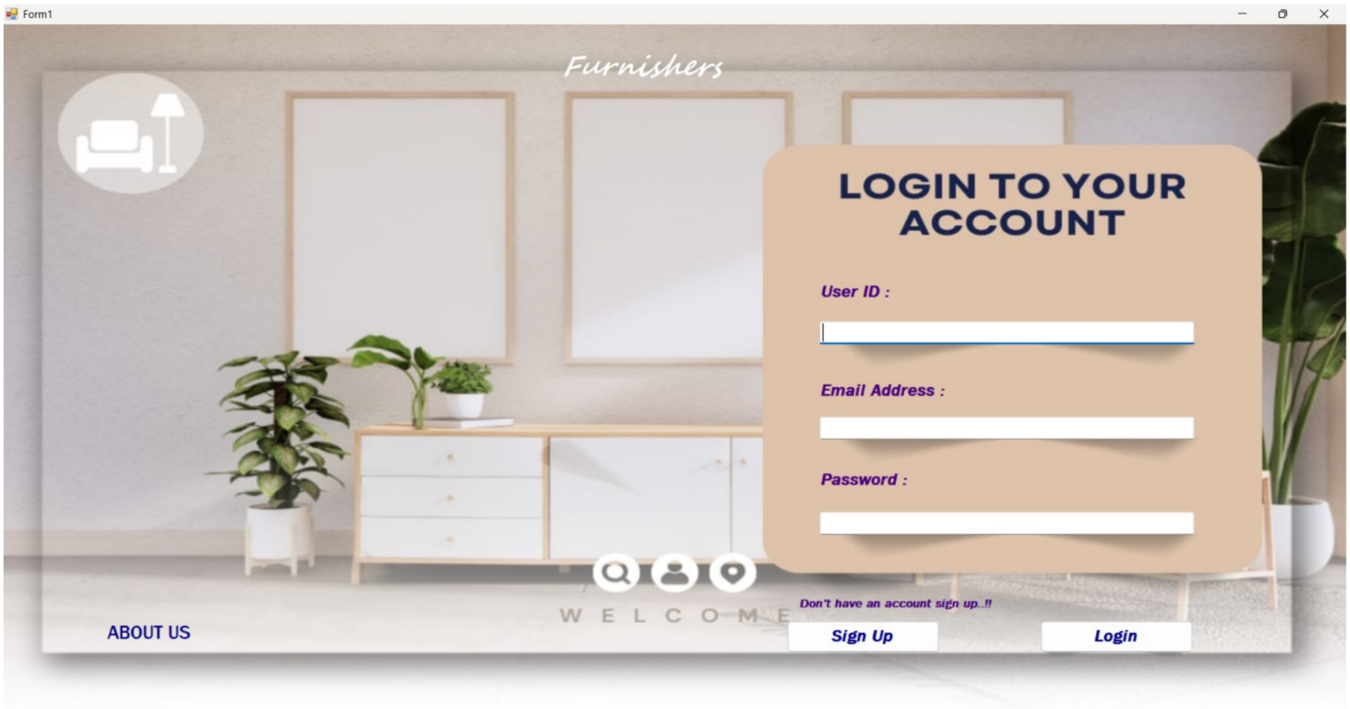
- **Login and Signup Screens:** Secure authentication interface for customers and admins.
- **Product Listing Pages:** Display categorized furniture items with search and filter options.
- **Cart and Checkout Screens:** Allow customers to review orders and proceed with payment.
- **Admin Dashboard:** Provides an overview of inventory, sales, and order status.
- **Billing Page:** Generates invoices and payment confirmations for completed purchases.

Process Involved

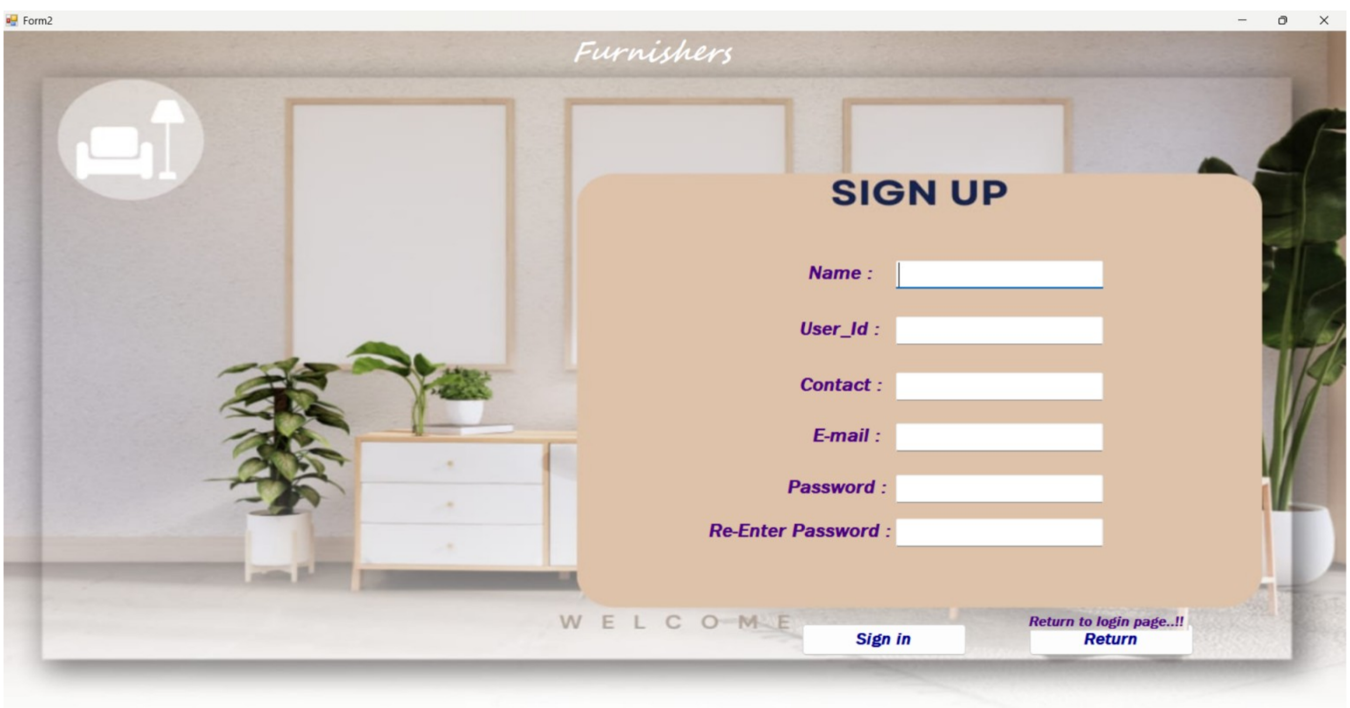
- **User Registration and Login:** Customers and admins create accounts and log in to access their respective functionalities.
- **Product Management:** Admins add, update, or remove furniture items.
- **Order Processing:** Customers place orders, which update inventory levels automatically.
- **Payment Handling:** Payments are processed, and invoices are generated.
- **Report Generation:** The system generates sales reports for business analysis.

USER SCREENS:

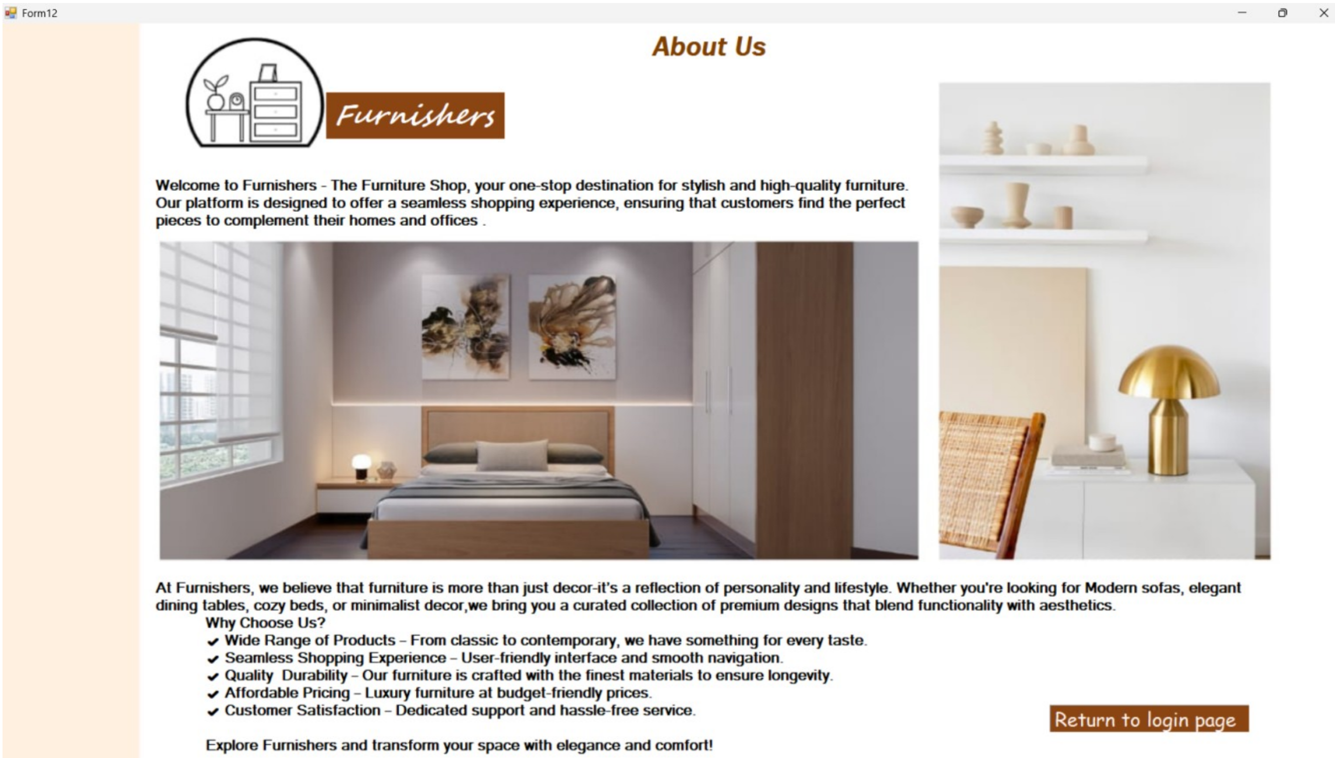
Login Page:-



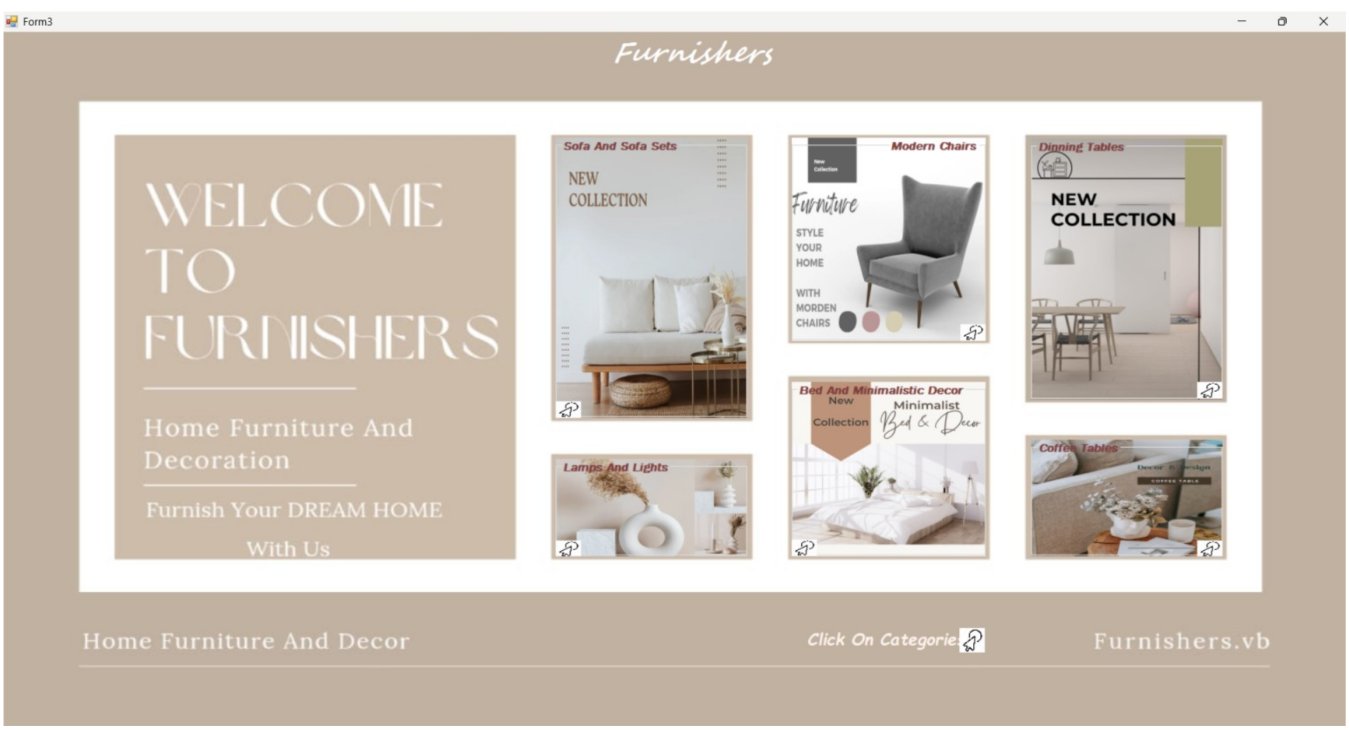
SIGNUP PAGE:-



ABOUT US PAGE:-



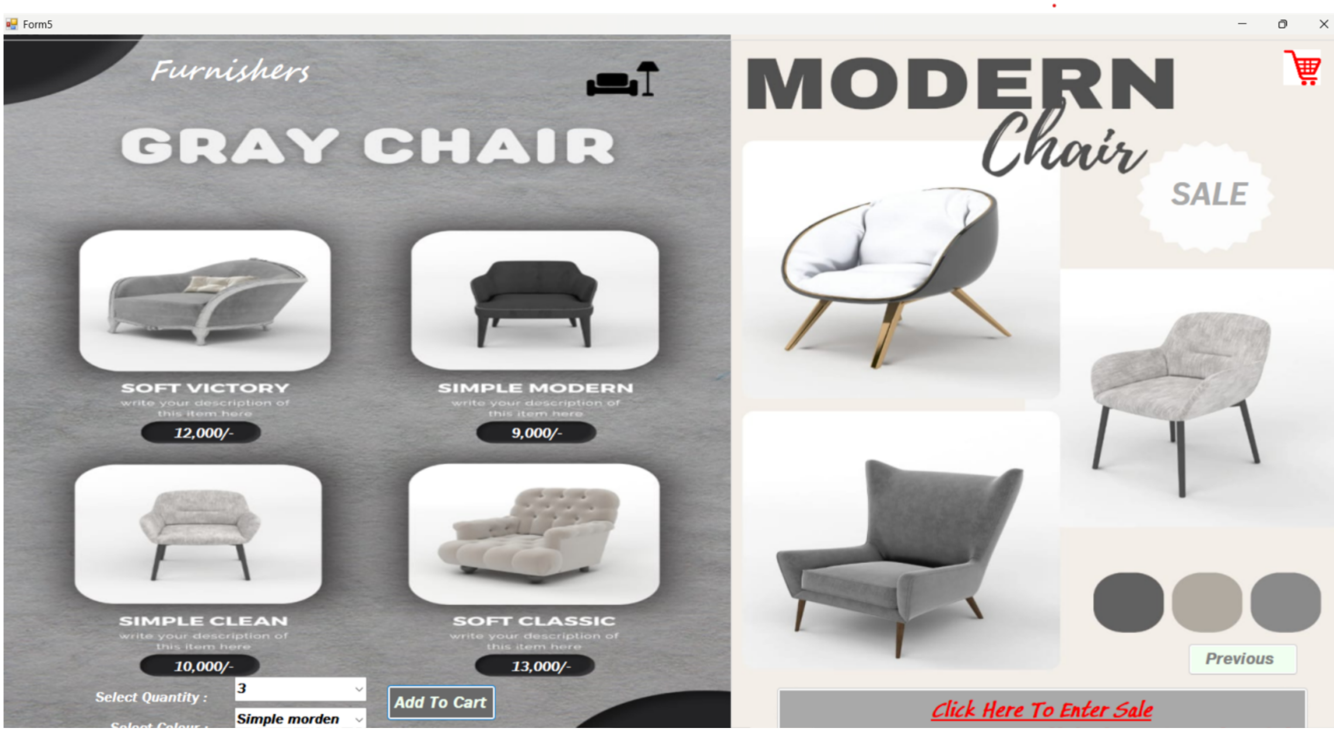
DASHBOARD PAGE:-



CATEGORY PAGES:-

1. **Sofa Catalogue**

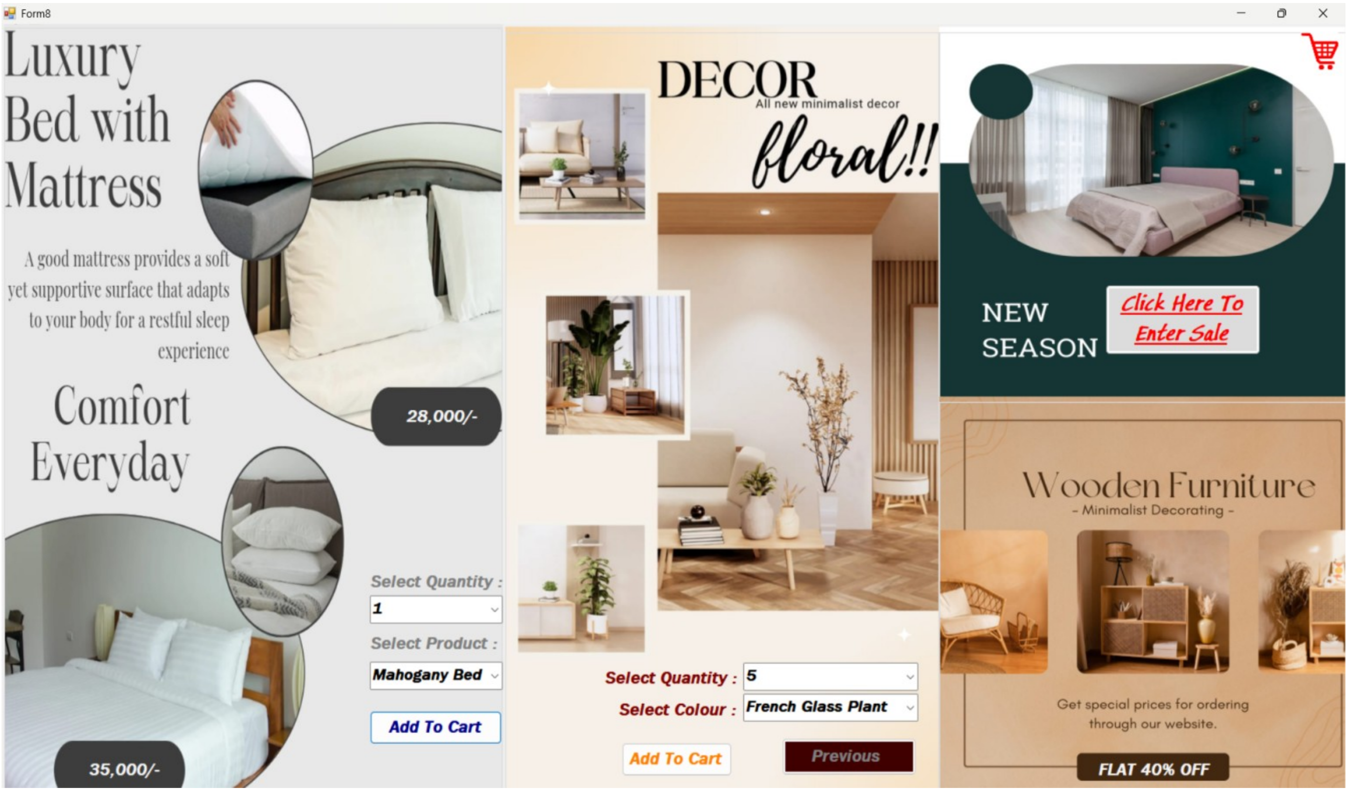
2. **Chair Catalogue**



3. Dinning Table Catalogue

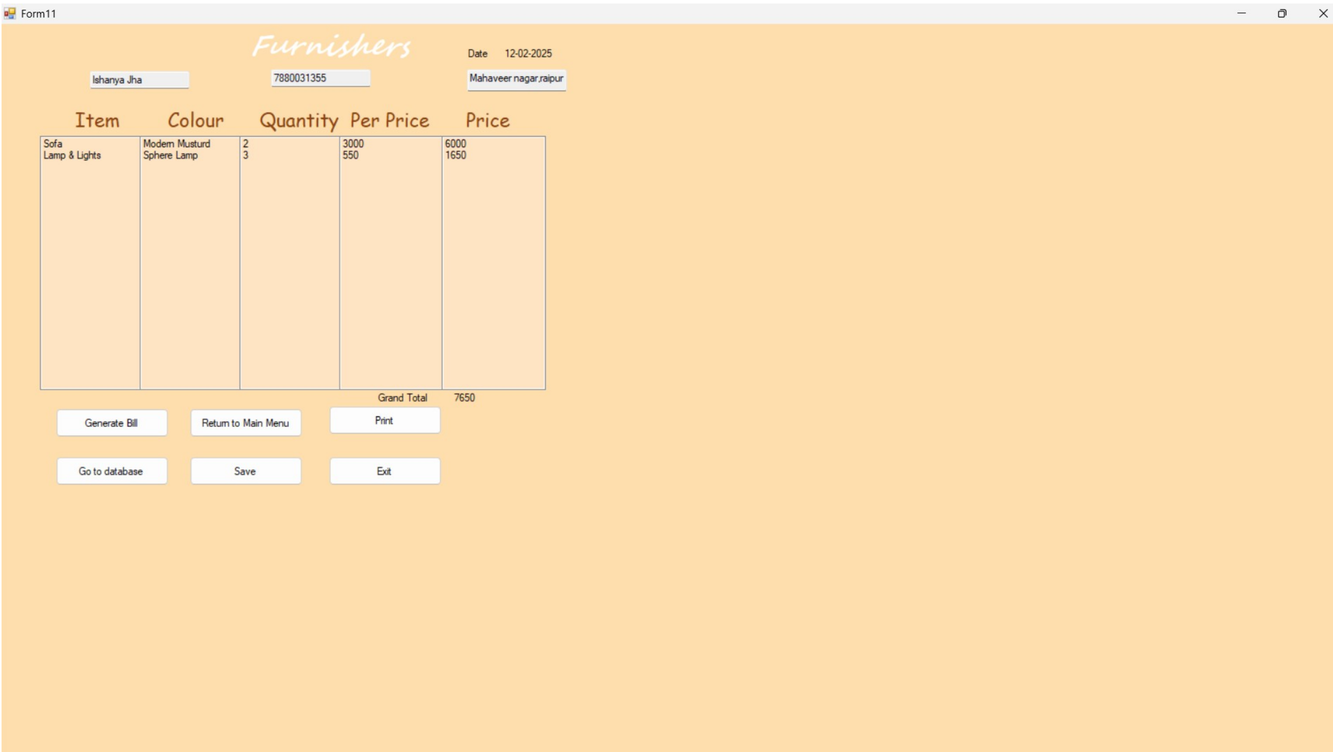
4. Lamp & Lights Catalogue

5. Bed & Minimalistic Decor Catalogue

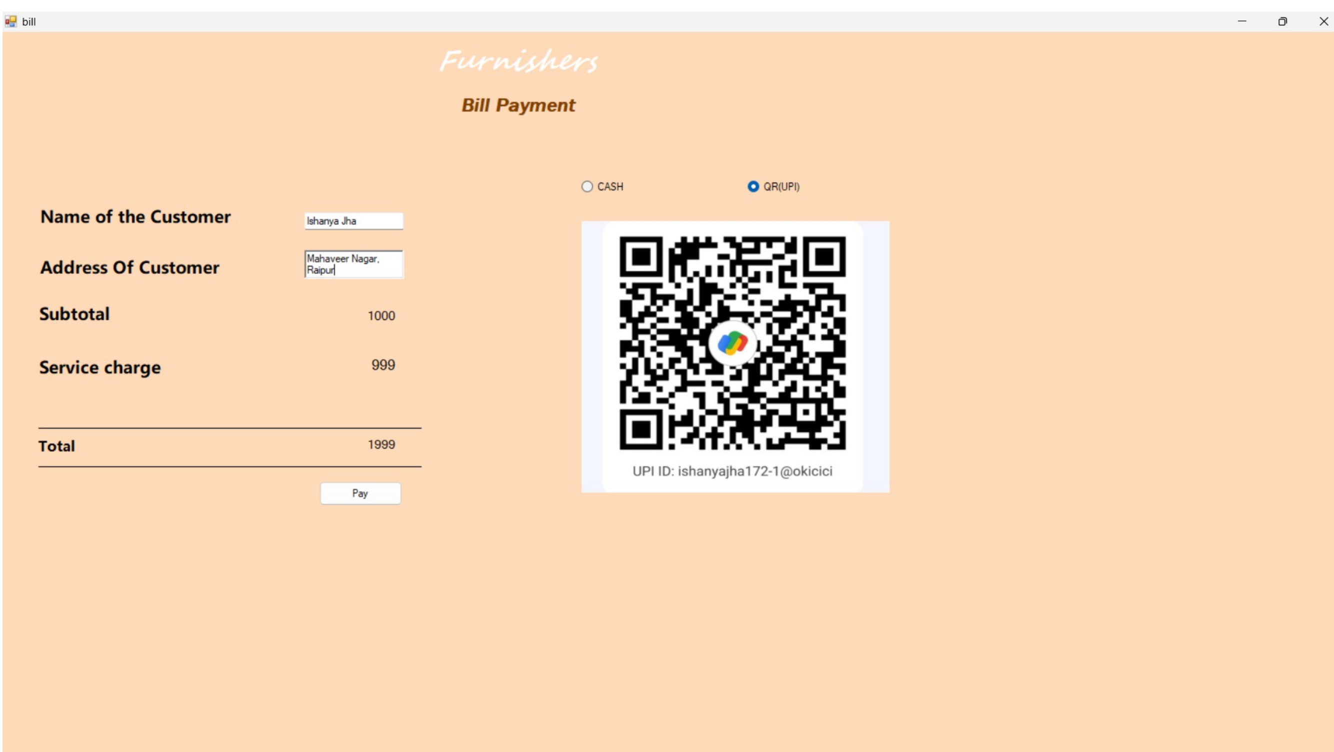


6. Coffee Table Catalogue

BILLING PAGE:-



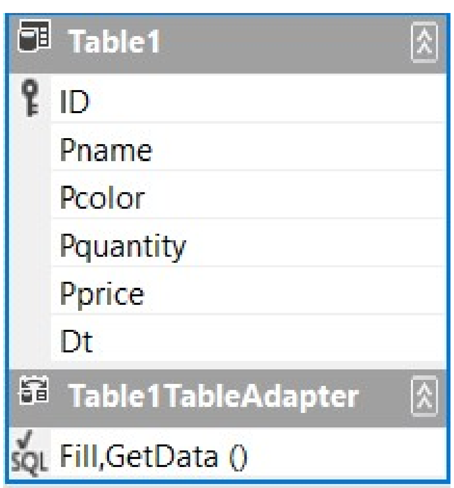
BILL PAYMENT PAGE:-



ADMIN SCREENS:

1. ADMIN PANEL

2. DATA SET



- **SOURCE CODE:-**

LOGIN PAGE:-

```
Public Class Form1

    Dim con As New OleDb.OleDbConnection("Provider=Microsoft.Jet.OLEDB.4.0;Data
Source=C:\Users\shara\OneDrive\Documents\Database1.mdb")

    Dim sql As String
    Dim cmd As New OleDb.OleDbCommand
    Dim dt As New DataTable
    Dim da As New OleDb.OleDbDataAdapter
    Dim i As Integer

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Me.Hide()
        Form2.Show()
    End Sub
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        Dim cmd As OleDb.OleDbCommand = New OleDb.OleDbCommand("SELECT  * FROM signup
WHERE [uid] = '" & TextBox1.Text & "'and [email]='" & TextBox2.Text & "' and [pass]='" &
TextBox3.Text & "' ", con)
        Dim user As String = ""
        Dim eml As String = ""
        Dim pass As String = ""
        Dim un As String = ""
      con.Open()
        Dim reader As OleDb.OleDbDataReader = cmd.ExecuteReader()

        If (reader.Read() = True) Then
            user = reader("uid")
            eml = reader("email")
            pass = reader("pass")
            un = reader("uname")
            Form11.TextBox1.Text = un
            MessageBox.Show("Login Successfully!")
            Dashboard.Show()
            Me.Hide()

        Else
            MsgBox("Invalid username and password!")
        End If

        reader.Close()
        con.Close()
```

```
    End Sub
    Private Sub Label6_Click(sender As Object, e As EventArgs) Handles Label6.Click
        AboutUs.Show()
        Me.Hide()
    End Sub

    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load
    End Sub

    Private Sub Label1_Click(sender As Object, e As EventArgs) Handles Label1.Click

    End Sub
End Class
```

SIGNUP PAGE:-

```
Public Class Form2
    Dim con As New OleDb.OleDbConnection("Provider=Microsoft.Jet.OLEDB.4.0;Data
Source=C:\Users\shara\OneDrive\Documents\Database1.mdb")

  Dim sql As String
  Dim cmd As New OleDb.OleDbCommand
  Dim dt As New DataTable
  Dim da As New OleDb.OleDbDataAdapter
  Dim i As Integer


  Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    Try

      con.Open()
      sql = "INSERT INTO signup ([uname],[uid],[cont],[email],[pass]) values ('" & TextBox1.Text &
"', '" & TextBox2.Text & "','" & TextBox3.Text & "','" & TextBox4.Text & "','" & TextBox5.Text & "')"

      cmd.Connection = con
      cmd.CommandText = sql
      i = cmd.ExecuteNonQuery
      If i > 0 Then
        MsgBox("Congratulations..!! register successful.")
      Else
        MsgBox("Please enter valid info.")
      End If
    Catch ex As Exception
      MsgBox(ex.Message)
    Finally
      con.Close()

    End Try

  End Sub

  Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Me.Hide()
    Form1.Show()
  End Sub

  Private Sub Form2_Load(sender As Object, e As EventArgs) Handles MyBase.Load

  End Sub
End Class
```

DASHBOARD PAGE:-

```
Public Class Dashboard

    Private Sub GroupBox1_Click(sender As Object, e As EventArgs) Handles GroupBox1.Click
        Form4.Show()
        Me.Hide()
    End Sub


    Private Sub GroupBox2_Click(sender As Object, e As EventArgs) Handles GroupBox2.Click
        ChairCatalogue.Show()
        Me.Hide()
    End Sub


    Private Sub GroupBox3_Click(sender As Object, e As EventArgs) Handles GroupBox3.Click
        DinningTableCatalogue.Show()
        Me.Hide()
    End Sub


    Private Sub GroupBox4_Click(sender As Object, e As EventArgs) Handles GroupBox4.Click
        LampsandLights.Show()
        Me.Hide()
    End Sub


    Private Sub GroupBox5_Click(sender As Object, e As EventArgs) Handles GroupBox5.Click
        BedandDecor.Show()
        Me.Hide()
    End Sub


    Private Sub GroupBox6_Click(sender As Object, e As EventArgs) Handles GroupBox6.Click
        Coffeetables.Show()
        Me.Hide()
    End Sub
    Private Sub Dashboard_Load(sender As Object, e As EventArgs) Handles MyBase.Load

    End Sub
End Class
```

CATEGORIES PAGES:-

1. Sofa Catalogue

```
Public Class Form4
    Dim st As Integer = 0
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        Dashboard.Show()
        Me.Hide()
    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        sale.Show()
        Me.Hide()
    End Sub

    Private Sub Form4_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ComboBox1.Items.Add("1")
        ComboBox1.Items.Add("2")
        ComboBox1.Items.Add("3")
        ComboBox1.Items.Add("4")
        ComboBox2.Items.Add("Beige")
        ComboBox2.Items.Add("Silver Moonlight")
        ComboBox2.Items.Add("Modern Musturd")
        ComboBox2.Items.Add("Off-White")
    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Form11.ListBox1.Items.Add("Sofa")
        Form11.ListBox2.Items.Add(ComboBox2.Text)
        Form11.ListBox3.Items.Add(ComboBox1.Text)

        If ComboBox2.Text = "Beige" Then
            st = Val(ComboBox1.Text) * 1000
            Form11.ListBox4.Items.Add("1000")
            total = total + st
        End If

        If ComboBox2.Text = "Silver Moonlight" Then
            st = Val(ComboBox1.Text) * 2000
            Form11.ListBox4.Items.Add("2000")
            total = total + st
        End If

        If ComboBox2.Text = "Modern Musturd" Then
            st = Val(ComboBox1.Text) * 3000
            Form11.ListBox4.Items.Add("3000")
```

```
            total = total + st
        End If

        If ComboBox2.Text = "Off-White" Then
            st = Val(ComboBox1.Text) * 4000
            Form11.ListBox4.Items.Add("4000")
            total = total + st
        End If

        Form11.ListBox5.Items.Add(st)
        count += 1
        MsgBox("Added to cart")
    End Sub

    Private Sub PictureBox4_Click(sender As Object, e As EventArgs) Handles PictureBox4.Click
        Form11.Show()
        Me.Hide()
    End Sub


End Class
```

2. Chair Catalogue

```
Public Class ChairCatalogue
    Dim st As Integer = 0
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        Dashboard.Show()
        Me.Hide()
    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        sale.Show()
        Me.Hide()
    End Sub

    Private Sub Form5_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ComboBox1.Items.Add("1")
        ComboBox1.Items.Add("2")
        ComboBox1.Items.Add("3")
        ComboBox1.Items.Add("4")
        ComboBox2.Items.Add("Simple morden ")
        ComboBox2.Items.Add("Soft classic ")
        ComboBox2.Items.Add("Soft victory ")
```

```
        ComboBox2.Items.Add("Simple clean")
    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Form11.ListBox1.Items.Add("Morden Chair")
        Form11.ListBox2.Items.Add(ComboBox2.Text)
        Form11.ListBox3.Items.Add(ComboBox1.Text)

        If ComboBox2.Text = "Simple morden" Then
            st = Val(ComboBox1.Text) * 2000
            Form11.ListBox4.Items.Add("2000")
            total = total + st
        End If

        If ComboBox2.Text = "Soft classic" Then
            st = Val(ComboBox1.Text) * 2000
            Form11.ListBox4.Items.Add("2000")
            total = total + st
        End If

        If ComboBox2.Text = "Soft victory" Then
            st = Val(ComboBox1.Text) * 3000
            Form11.ListBox4.Items.Add("3000")
            total = total + st
        End If

        If ComboBox2.Text = "Simple clean" Then
            st = Val(ComboBox1.Text) * 4000
            Form11.ListBox4.Items.Add("4000")
            total = total + st
        End If

        Form11.ListBox5.Items.Add(st)
        count += 1
        MsgBox("Added to cart")
    End Sub


    Private Sub PictureBox4_Click(sender As Object, e As EventArgs) Handles PictureBox4.Click
        Form11.Show()
        Me.Hide()
    End Sub

End Class
```

3. Dinning Table Catalogue

```
Public Class DinningTableCatalogue
    Dim st As Integer = 0
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        sale.Show()
        Me.Hide()
    End Sub
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        Dashboard.Show()
        Me.Hide()
    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Form11.ListBox1.Items.Add("Bed & Decor")
        Form11.ListBox2.Items.Add(ComboBox2.Text)
        Form11.ListBox3.Items.Add(ComboBox1.Text)

        If ComboBox2.Text = "Beige" Then
            st = Val(ComboBox1.Text) * 1000
            Form11.ListBox4.Items.Add("1000")
            total = total + st
        End If
        If ComboBox2.Text = "Silver Moonlight" Then
            st = Val(ComboBox1.Text) * 2000
            Form11.ListBox4.Items.Add("2000")
            total = total + st
        End If
        If ComboBox2.Text = "Modern Musturd" Then
            st = Val(ComboBox1.Text) * 3000
            Form11.ListBox4.Items.Add("3000")
            total = total + st
        End If
        If ComboBox2.Text = "Off-White" Then
            st = Val(ComboBox1.Text) * 4000
            Form11.ListBox4.Items.Add("4000")
            total = total + st
        End If
        Form11.ListBox5.Items.Add(st)
        count += 1
        MsgBox("Added to cart")
    End Sub

    Private Sub PictureBox4_Click(sender As Object, e As EventArgs) Handles PictureBox4.Click
        Form11.Show()
        Me.Hide()
    End Sub
    Private Sub ComboBox2_SelectedIndexChanged(sender As Object, e As EventArgs) Handles
```

```
ComboBox2.SelectedIndexChanged
    End Sub

    Private Sub Form6_Load(sender As Object, e As EventArgs) Handles MyBase.Load

    End Sub
End Class
```

4. Lamp & Lights Catalogue

```
Public Class LampsandLights
    Dim st As Integer = 0
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        Dashboard.Show()
        Me.Hide()
    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        sale.Show()
        Me.Hide()
    End Sub

    Private Sub Form5_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ComboBox1.Items.Add("1")
        ComboBox1.Items.Add("2")
        ComboBox1.Items.Add("3")
        ComboBox1.Items.Add("4")
        ComboBox2.Items.Add("Silver Moonlight")
        ComboBox2.Items.Add("Modern Musturd")
        ComboBox2.Items.Add("Off-White")
    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Form11.ListBox1.Items.Add("Lamp & Lights")
        Form11.ListBox2.Items.Add(ComboBox2.Text)
        Form11.ListBox3.Items.Add(ComboBox1.Text)

        If ComboBox2.Text = "Beige" Then
            st = Val(ComboBox1.Text) * 1000
            Form11.ListBox4.Items.Add("1000")
            total = total + st
        End If
        If ComboBox2.Text = "Silver Moonlight" Then
            st = Val(ComboBox1.Text) * 2000
            Form11.ListBox4.Items.Add("2000")
            total = total + st
```

```
        End If
        If ComboBox2.Text = "Modern Musturd" Then
            st = Val(ComboBox1.Text) * 3000
            Form11.ListBox4.Items.Add("3000")
            total = total + st
        End If
        If ComboBox2.Text = "Off-White" Then
            st = Val(ComboBox1.Text) * 4000
            Form11.ListBox4.Items.Add("4000")
            total = total + st
        End If
        Form11.ListBox5.Items.Add(st)
        count += 1
        MsgBox("Added to cart")
    End Sub

    Private Sub PictureBox4_Click(sender As Object, e As EventArgs) Handles PictureBox4.Click
        Form11.Show()
        Me.Hide()
    End Sub
End Class
```

5. Bed & Minimalistic Decor Catalogue

```
Public Class BedandDecor
    Dim st As Integer = 0
    Dim s As Integer = 0
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        Dashboard.Show()
        Me.Hide()
    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        sale.Show()
        Me.Hide()
    End Sub


    Private Sub Form5_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ComboBox1.Items.Add("1")
        ComboBox1.Items.Add("2")
        ComboBox1.Items.Add("3")
        ComboBox1.Items.Add("4")
        ComboBox2.Items.Add("Sleepwell Grey Bed")
        ComboBox2.Items.Add("Mahogany Wooden Bed")
        ComboBox4.Items.Add("1")
```

```
    ComboBox4.Items.Add("2")
    ComboBox4.Items.Add("3")
    ComboBox4.Items.Add("4")
    ComboBox3.Items.Add("French Grass Plant")
    ComboBox3.Items.Add("German Grass Plant")
    ComboBox3.Items.Add("Philodendron Grass Plant")
    ComboBox3.Items.Add("Paperonia Grass Plant")

End Sub

Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Form11.ListBox1.Items.Add("Bed with Mattress")
    Form11.ListBox2.Items.Add(ComboBox2.Text)
    Form11.ListBox3.Items.Add(ComboBox1.Text)

    If ComboBox2.Text = "Sleepwell Grey Bed" Then
        st = Val(ComboBox1.Text) * 28000
        Form11.ListBox4.Items.Add("28000")
        total = total + st
    End If
    If ComboBox2.Text = "Mahogany Wooden Bed" Then
        st = Val(ComboBox1.Text) * 35000
        Form11.ListBox4.Items.Add("35000")
        total = total + st
    End If

    Form11.ListBox5.Items.Add(st)
    count += 1
    MsgBox("Added to cart")
End Sub

Private Sub PictureBox4_Click(sender As Object, e As EventArgs) Handles PictureBox4.Click
    Form11.Show()
    Me.Hide()
End Sub


Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
    Form11.ListBox1.Items.Add("Plants")
    Form11.ListBox2.Items.Add(ComboBox3.Text)
    Form11.ListBox3.Items.Add(ComboBox4.Text)

    If ComboBox3.Text = "French Grass Plant" Then
        s = Val(ComboBox4.Text) * 2800
        Form11.ListBox4.Items.Add("2800")
        total = total + s
    End If
    If ComboBox3.Text = "German Grass Plant" Then
```

```
            s = Val(ComboBox4.Text) * 350
            Form11.ListBox4.Items.Add("350")
            total = total + s
        End If

        If ComboBox3.Text = "Philodendron Grass Plant" Then
            s = Val(ComboBox4.Text) * 2800
            Form11.ListBox4.Items.Add("2800")
            total = total + s
        End If
        If ComboBox3.Text = "Paperonia Grass Plant" Then
            s = Val(ComboBox4.Text) * 350
            Form11.ListBox4.Items.Add("350")
            total = total + s
        End If
        Form11.ListBox5.Items.Add(s)
        count += 1
        MsgBox("Added to cart")
    End Sub
End Class
```

6. Coffee Table Catalogue

```
Public Class Coffeetables

    Dim st As Integer = 0
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        Dashboard.Show()
        Me.Hide()
    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        sale.Show()
        Me.Hide()
    End Sub


    Private Sub Form5_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ComboBox1.Items.Add("1")
        ComboBox1.Items.Add("2")
        ComboBox1.Items.Add("3")
        ComboBox1.Items.Add("4")
        ComboBox2.Items.Add("Biege")
        ComboBox2.Items.Add("Silver Moonlight")
        ComboBox2.Items.Add("Modern Musturd")
        ComboBox2.Items.Add("Off-White")
```

```
    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Form11.ListBox1.Items.Add("Coffee Table")
        Form11.ListBox2.Items.Add(ComboBox2.Text)
        Form11.ListBox3.Items.Add(ComboBox1.Text)

        If ComboBox2.Text = "Beige" Then
            st = Val(ComboBox1.Text) * 1000
            Form11.ListBox4.Items.Add("1000")
            total = total + st
        End If
        If ComboBox2.Text = "Silver Moonlight" Then
            st = Val(ComboBox1.Text) * 2000
            Form11.ListBox4.Items.Add("2000")
            total = total + st
        End If
        If ComboBox2.Text = "Modern Musturd" Then
            st = Val(ComboBox1.Text) * 3000
            Form11.ListBox4.Items.Add("3000")
            total = total + st
        End If
        If ComboBox2.Text = "Off-White" Then
            st = Val(ComboBox1.Text) * 4000
            Form11.ListBox4.Items.Add("4000")
            total = total + st
        End If
        Form11.ListBox5.Items.Add(st)
        count += 1
        MsgBox("Added to cart")
    End Sub

    Private Sub PictureBox4_Click(sender As Object, e As EventArgs) Handles PictureBox4.Click
        Form11.Show()
        Me.Hide()
    End Sub
End Class
```

BILLING PAGE:-

```
Public Class Form11
    Dim con As New OleDb.OleDbConnection("Provider=Microsoft.Jet.OLEDB.4.0;Data
Source=C:\Users\shara\OneDrive\Documents\Database1.mdb")
    Dim sql As String
    Dim cmd As New OleDb.OleDbCommand
    Dim dt As New DataTable
    Dim da As New OleDb.OleDbDataAdapter
    Dim i As Integer

    Private Sub Form11_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ' TextBox3.Text = bill.RichTextBox1.Text
        Label15.Text = Date.Today
        Label10.Text = total

        Label16.Hide()
        Label17.Hide()
        Label8.Hide()

    End Sub

    Private Sub Label12_Click(sender As Object, e As EventArgs)
        AdminPanel.Show()
        Me.Hide()
    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click

        'If TextBox3.Text = "" Then
        'MsgBox("Enter Delivery Address")
        'Else

        bill.Show()
        bill.Activate()

        Me.Hide()
        Button1.Visible = False

        ' TextBox1.Show()

        'RichTextBox1.Show()
        'Label16.Show()
        'Label17.Show()

        ' bill.TextBox1.Text = TextBox1.Text
        ' End If
    End Sub
```

```
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        Me.Hide()
        Dashboard.Show()
    End Sub

    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        AdminPanel.Show()
        Me.Hide()
    End Sub

    Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
        PrintForm1.PrintAction = Printing.PrintAction.PrintToPreview
        PrintForm1.Print()
    End Sub

    Private Sub Button5_Click(sender As Object, e As EventArgs) Handles Button5.Click

        Try
            con.Open()
            For j As Integer = 0 To (count - 1)

                sql = "INSERT INTO Table1 (cname,caddress,Pname,Pcolor,Pquantity,Pprice,Dt) values ('" &
TextBox1.Text & "','" & TextBox3.Text & "','" & ListBox1.Items(j) & "', '" & ListBox2.Items(j) & "','"
& ListBox3.Items(j) & "','" & ListBox5.Items(j) & "','" & Label15.Text & "')"
            Next

            cmd.Connection = con
            cmd.CommandText = sql
            i = cmd.ExecuteNonQuery
            If i > 0 Then
                MsgBox("New record has been inserted successfully!")
            Else
                MsgBox("No record has been inserted successfully!")
            End If

        Catch ex As Exception
            MsgBox(ex.Message)
        Finally
            con.Close()

        End Try
    End Sub

    Private Sub Button6_Click(sender As Object, e As EventArgs) Handles Button6.Click
        Me.Close()
    End Sub
End Clas
```

BILL PAYMENT PAGE:-

```
Public Class bill


    Private Sub bill_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        Label4.Text = total
        Label6.Text = total + 999
    End Sub
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        If RadioButton1.Checked Then
            MsgBox("Payment Successful.")
            Form11.Show()
            Me.Hide()
        ElseIf RadioButton2.Checked Then
            MsgBox("Payment Successful.")
            Form11.Show()
            Me.Hide()
        Else
            MsgBox("Please select payment method.")
        End If


    End Sub
    Private Sub RadioButton1_CheckedChanged(sender As Object, e As EventArgs) Handles
RadioButton1.CheckedChanged
        If RadioButton1.Checked Then
            PictureBox1.Load("C:\Users\shara\Downloads\Cash.png")
        End If
        If RadioButton2.Checked Then
            PictureBox1.Load("C:\Users\shara\Downloads\Qr.jpg")
        End If
    End Sub

    Private Sub PictureBox1_Click(sender As Object, e As EventArgs) Handles PictureBox1.Click

    End Sub
End Class
```

ADMIN :

1. ADMIN PANEL :-

```
Public Class AdminPanel
    Dim con As New OleDb.OleDbConnection("Provider=Microsoft.Jet.OLEDB.4.0;Data
Source=C:\Users\shara\OneDrive\Documents\Database1.mdb")
    Dim sql As String
    Dim cmd As New OleDb.OleDbCommand
    Dim dt As New DataTable
    Dim da As New OleDb.OleDbDataAdapter
    Dim i As Integer

    Private Sub Form13_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        'TODO: This line of code loads data into the 'DataSet1.Table1' table. You can move, or remove it,
as needed.
        Me.Table1TableAdapter.Fill(Me.DataSet1.Table1)

        Try

            con.Open()
            sql = "Select * from Table1"
            cmd.Connection = con
            cmd.CommandText = sql
            da.SelectCommand = cmd

            da.Fill(dt)

            DataGridView1.DataSource = dt
        Catch ex As Exception
            MsgBox(ex.Message)
        Finally
            con.Close()

        End Try


    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Form11.Show()
        Me.Hide()
    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        Dim selectString As String = "SELECT  *FROM Table1  WHERE ID = " & Val(TextBox1.Text)
& ""
```

```
        con.Open()

        Dim cmd As OleDb.OleDbCommand = New OleDb.OleDbCommand(selectString, con)


        Dim reader As OleDb.OleDbDataReader = cmd.ExecuteReader()



        reader.Read()
        TextBox1.Text = reader.GetValue(0).ToString()
        TextBox2.Text = reader.GetValue(1).ToString()
        TextBox3.Text = reader.GetValue(2).ToString()
        TextBox4.Text = reader.GetValue(3).ToString()
        TextBox5.Text = reader.GetValue(4).ToString()
        TextBox6.Text = reader.GetValue(5).ToString()
        TextBox7.Text = reader.GetValue(6).ToString()
        TextBox8.Text = reader.GetValue(7).ToString()

        reader.Close()
        con.Close()

    End Sub
End Class
```

2. APP.Config

```
<?xml version="1.0" encoding="utf-8" ?>

<configuration>

    <configSections>

    </configSections>

    <connectionStrings>

        <add name="WindowsApplication1.My.MySettings.ConnectionString"
            connectionString="Dsn=fur" providerName="System.Data.Odbc" />

    </connectionStrings>

    <startup>

        <supportedRuntime version="v4.0" sku=".NETFramework,Version=v4.5" />

    </startup>

</configuration>
```

METHODOLOGY USED FOR TESTING

The application undergoes multiple testing phases to ensure functionality, security, and performance.

- **Unit Testing:** Individual components like login, product search, and checkout are tested separately.
- **Integration Testing:** Ensures smooth interaction between modules such as inventory management and billing.
- **User Acceptance Testing (UAT):** The system is tested in a real-world environment to ensure usability and reliability.

Test Report and Code Printout:

All test cases are documented, detailing expected vs. actual results. A code printout is maintained for reference, covering key modules such as authentication, product management, and billing.

The **Furnishers - The Furniture Shop** project follows a well-defined **software development life cycle (SDLC)**, ensuring a robust and efficient furniture store management system.

USER AND OPERATIONAL MANUAL

The Furnishers - The Furniture Shop application provides an intuitive interface for both customers and administrators. This manual guides users through the system's functionalities and security aspects to ensure proper usage and data protection.

Customer Operations:

- **Login and Signup:** Customers must register an account using their email and password. Returning users can log in securely.
- **Product Browsing:** Customers can view different furniture categories, search for specific items, and filter products based on price and availability.
- **Cart Management:** Selected products are added to the cart for checkout. Users can modify item quantities or remove items before proceeding.
- **Order Placement:** Customers confirm their purchase by completing the checkout process. The system generates an invoice upon successful payment.
- **Payment Processing:** The system provides multiple payment options. Once a transaction is completed, the order is stored in the database.

Admin Operations:

- **User Authentication:** Admins log in with their credentials to access the management dashboard.
- **Inventory Management:** Admins can add, update, or remove furniture items, ensuring the catalog remains up to date.
- **Order Tracking:** The dashboard displays all customer orders, including pending, processed, and completed transactions.
- **Billing and Reports:** The system automatically generates invoices for purchases and provides sales reports for business insights.

Security Aspects:

- **Role-Based Access Control (RBAC):** The system ensures that only admins can modify inventory, track orders, and generate reports, while customers can only browse and purchase products.
- **Secure Authentication:** Password encryption and session management prevent unauthorized access.
- **Data Encryption:** Sensitive customer data, including login credentials and billing details, are encrypted for security.
- **Audit Logs:** The system records login attempts, inventory changes, and order updates to maintain security and prevent misuse.

Backup and Data Recovery:

- **Automated Backups:** The database automatically backs up product details, order history, and user accounts at regular intervals.
- **Manual Backup Option:** Admins can create manual backups of inventory and sales records for additional data protection.
- **Recovery Mechanism:** In case of data loss, backup files can be restored to maintain business continuity.

System Controls and Validation

- **Form Validations:** Input fields for login, product management, and checkout include validation checks to prevent incorrect or incomplete entries.
- **Error Handling:** The system provides error messages and prompts in case of invalid operations, preventing crashes and data inconsistencies.
- **Transaction Management:** Orders and payments follow secure transaction protocols, ensuring successful completion without duplication or loss of data.

By implementing strong **security, backup, and control measures**, the **Furnishers - The Furniture Shop** system ensures safe, efficient, and reliable furniture store management, protecting both business and customer data.

CONCLUSION

The **Furnishers - The Furniture Shop** project successfully addresses the challenges faced by traditional furniture stores by providing a **digitized and automated** solution for managing inventory, orders, and billing. Developed using **VB.NET** with **MS Access** as the database and **OLEDB connectivity**, the system ensures a **structured, efficient, and user-friendly platform** for both customers and administrators.

By integrating key functionalities such as secure authentication, product browsing, cart management, and automated invoicing, the application enhances the customer experiencewhile streamlining store operations. The **admin panel** provides complete control over inventory management, order processing, and sales tracking, allowing business owners to make **data-driven decisions** and optimize store performance.

The implementation of security measures, including role-based access control, encrypted transactions, and automated backups, ensures data integrity, reliability, and protection against unauthorized access.

Additionally, the system's **scalability** allows for future enhancements such as advanced reporting, multi-store integration, and online order tracking.

With the successful development and deployment of this application, furniture store owners can experience increased efficiency, reduced manual errors, and improved customer satisfaction. The **Furnishers - The Furniture Shop** system is a comprehensive and modernized approach to furniture retail management, ensuring stainable business growth and operational excellence in the long run.

References

- **VB.NET Documentation:** Official Microsoft documentation for Windows Forms and OLEDB connectivity.
- **MS Access Database:** Used for storing inventory, orders, and user data.
- **Visual Studio:** Primary Integrated Development Environment (IDE) for coding, debugging, and testing.
- **Canva :** Used for creating wireframes and UI/UX design concepts.
- **Online Tutorials and Courses:** References from online learning platforms such as W3Schools, GeeksforGeeks, and YouTube tutorials for implementing features in VB.NET.

These references contributed to the successful development of a **structured, functional, and visually appealing** furniture shop management system.